

# Introduction to Social Data Analytics: Week 9 Part 1

# U.S. 2000 Presidential Election

- ▶ George Bush vs. Al Gore
- ▶ Bush: 271 electoral votes, Gore: 266
- ▶ Very close election, long recount

## Controversy in Florida

- ▶ Florida, 25 electoral votes, was particularly close
  - ▶ In the end, Bush won Florida by just 537 votes!
- ▶ Of particular controversy: butterfly ballot in Palm Beach County
- ▶ Many voters complained that they had intended to vote for Gore, but had actually voted for Pat Buchanan (Reform Party).

# Butterfly Ballot

| PALM BEACH COUNTY, FLORIDA<br>NOVEMBER 7, 2000 |                                |      | PALM BEACH COUNTY, FLORIDA<br>NOVEMBER 7, 2000                                                                         |                               |                                                                                     |
|------------------------------------------------|--------------------------------|------|------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------|
| es will<br>electors.)                          | (REPUBLICAN)                   |      |                                                                                                                        |                               |                                                                                     |
|                                                | GEORGE W. BUSH - PRESIDENT     | 3 ➡  |                                                                                                                        |                               |                                                                                     |
|                                                | DICK CHENEY - VICE PRESIDENT   |      |                                                                                                                        |                               |                                                                                     |
|                                                | (DEMOCRATIC)                   |      | 4 ←                                                                                                                    | (REFORM)                      |                                                                                     |
|                                                | AL GORE - PRESIDENT            | 5 ➡  |                                                                                                                        | PAT BUCHANAN - PRESIDENT      |                                                                                     |
|                                                | JOE LIEBERMAN - VICE PRESIDENT |      |                                                                                                                        | EZOLA FOSTER - VICE PRESIDENT |                                                                                     |
|                                                | (LIBERTARIAN)                  |      | 6 ←                                                                                                                    | (SOCIALIST)                   |                                                                                     |
| HARRY BROWNE - PRESIDENT                       | 7 ➡                            |      | DAVID McREYNOLDS - PRESIDENT                                                                                           |                               |                                                                                     |
| ART OLIVIER - VICE PRESIDENT                   |                                |      | MARY CAL HOLLIS - VICE PRESIDENT                                                                                       |                               |                                                                                     |
| (GREEN)                                        |                                | 8 ←  | (CONSTITUTION)                                                                                                         |                               |  |
| RALPH NADER - PRESIDENT                        | 9 ➡                            |      | HOWARD PHILLIPS - PRESIDENT                                                                                            |                               |                                                                                     |
| WINONA LaDUKE - VICE PRESIDENT                 |                                |      | J. CURTIS FRAZIER - VICE PRESIDENT                                                                                     |                               |                                                                                     |
| (SOCIALIST WORKERS)                            |                                | 10 ← | (WORKERS WORLD)                                                                                                        |                               |                                                                                     |
| JAMES HARRIS - PRESIDENT                       | 11 ➡                           |      | MONICA MOOREHEAD - PRESIDENT                                                                                           |                               |                                                                                     |
| MARGARET TROWE - VICE PRESIDENT                |                                |      | GLORIA La RIVA - VICE PRESIDENT                                                                                        |                               |                                                                                     |
| (NATURAL LAW)                                  |                                |      |                                                                                                                        |                               |                                                                                     |
| JOHN HAGELIN - PRESIDENT                       | 13 ➡                           |      |                                                                                                                        |                               |                                                                                     |
| NAT GOLDBABER - VICE PRESIDENT                 |                                |      |                                                                                                                        |                               |                                                                                     |
|                                                |                                |      | WRITE-IN CANDIDATE<br>To vote for a write-in candidate, follow the<br>directions on the long stub of your ballot card. |                               |                                                                                     |

## Palm Beach County

- ▶ Wand et al (2001) “The Butterfly Did It: The Aberrant Vote for Buchanan in Palm Beach County, Florida” *American Political Science Review*
- ▶ Use votes from 1996 U.S. Presidential Election to predict those in 2000.
- ▶ Specifically, use votes in each county for Ross Perot (Reform Party) in 1996 to predict votes for Buchanan in 2000.
- ▶ Was Palm Beach County an outlier?

## Load the data

```
florida <- read.csv("florida.csv")  
head(florida)
```

```
##      county Clinton96 Dole96 Perot96 Bush00 Gore00 Buchanan00  
## 1  Alachua      40144  25303    8072  34124  47365          263  
## 2   Baker       2273   3684     667   5610   2392           73  
## 3    Bay       17020  28290    5922  38637  18850          248  
## 4 Bradford     3356   4038     819   5414   3075           65  
## 5  Brevard     80416  87980   25249 115185  97318          570  
## 6  Broward    320736 142834   38964 177323 386561          788
```

```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.1.2
```

```
## Warning: package 'ggplot2' was built under R version 4.1.2
```

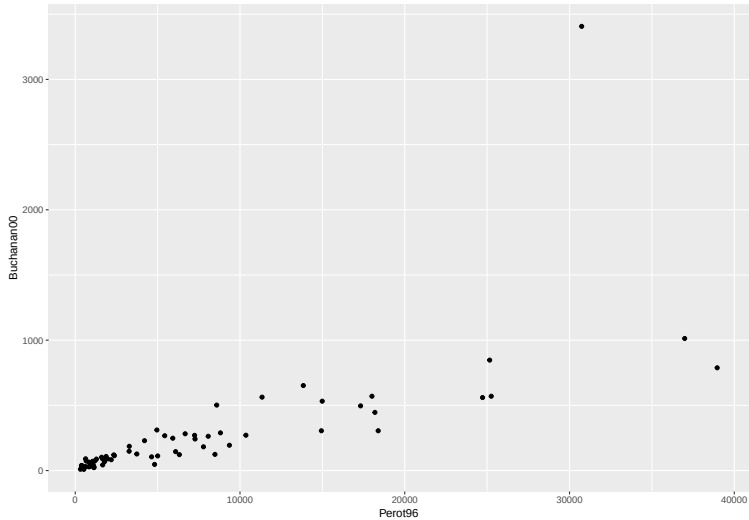
```
## Warning: package 'tibble' was built under R version 4.1.2
```

```
## Warning: package 'tidyr' was built under R version 4.1.2
```

## Plot votes for Perot and votes for Buchanan

```
ggplot(florida, mapping=aes(x=Perot96, y=Buchanan00)) +  
  geom_point()
```

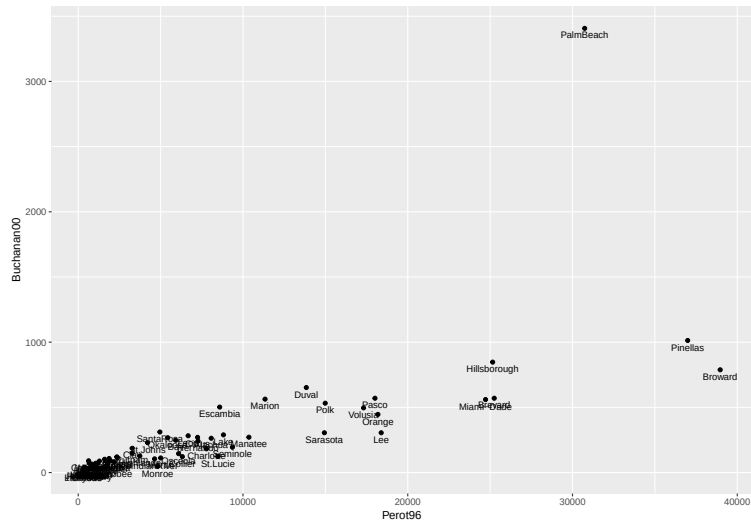
## Plot votes for Perot and votes for Buchanan



## Add label text of county

```
ggplot(florida, mapping=aes(x=Perot96, y=Buchanan00,  
  label=county)) +  
  geom_point() +  
  geom_text(vjust=1.5, size=3)
```

# Add label text of county



## Review: Linear Regression

We can write the linear regression model as:

$$Y_i = \beta_0 + \beta_1 \cdot X_i + \varepsilon_i$$

- ▶  $Y_i$ : dependent variable
- ▶  $X_i$ : independent variable
- ▶  $\beta_0$ : intercept
- ▶  $\beta_1$ : slope
- ▶  $\varepsilon_i$ : error

## Review: Linear Regression

We fit the line by minimizing the sum of squared errors:

- ▶ Find  $\hat{\beta}_0$  and  $\hat{\beta}_1$  that minimizes the sum of the squared errors,  $\sum_{i=1}^N (Y_i - \hat{Y}_i)^2$

Interpretation:

- ▶ A one unit increase in  $X$  is associated with  $\hat{\beta}_1$ -unit change in  $Y$ .

## Prediction Buchanan00

- ▶ Let's predict the number of votes in a county for Buchanan in 2000, based on the number of votes for Perot in 1996.

$$Buchanan00_i = \beta_0 + \beta_1 \cdot Perot96_i + \varepsilon_i$$

- ▶ Then we can form predicted values for each county
- ▶ If the predicted value for Palm Beach is far from the actual value, then something unusual is happening in Palm Beach

## lm(function)

- ▶ The function you use to estimate linear regression in R is the `lm()` function (which stands for linear model). The general syntax is

```
lm.fit <- lm(yvar ~ xvar, data=df)
```

- ▶ `lm.fit` stores the estimates of the linear regression as an object in R
- ▶ `yvar` should be replaced with the **dependent** variable
- ▶ `xvar` should be replaced with the **independent** variable
- ▶ `df` should be replaced with whatever the dataframe is named in your environment

## Estimate regression predicting Buchanan00 with Perot96

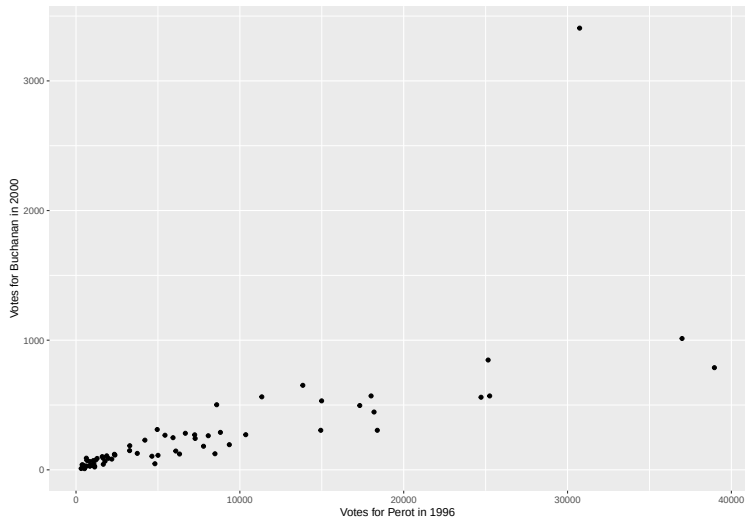
```
lm.florida <- lm(Buchanan00 ~ Perot96, data=florida)
lm.florida

##
## Call:
## lm(formula = Buchanan00 ~ Perot96, data = florida)
##
## Coefficients:
## (Intercept)      Perot96
##      1.34575      0.03592
```

## Interpretation

- ▶ We found  $\hat{\beta}_1 = 0.036$ 
  - ▶ Implies an additional vote for Perot in 1996 is predicted to increase votes for Buchanan by 0.036
- ▶ Now let's learn how to add a linear regression line to our scatterplot

## Plot votes for Perot and votes for Buchanan in ggplot



## geom\_smooth

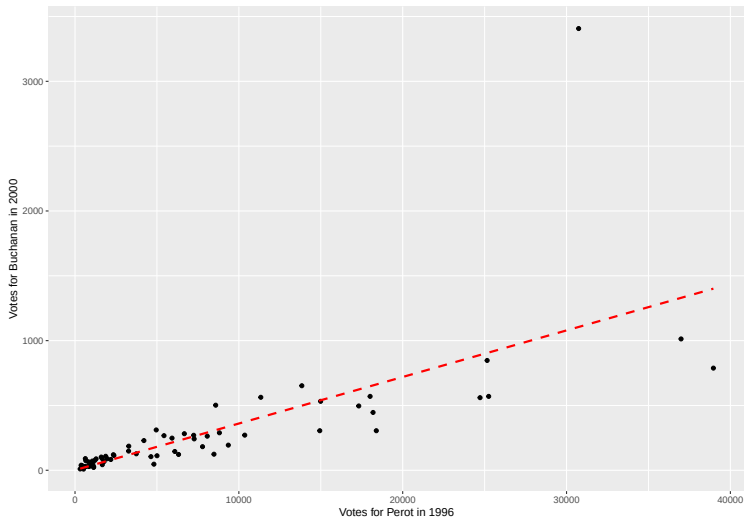
- ▶ To add a regression line we can use the `geom_smooth` option in `ggplot`
- ▶ To add a linear regression line we specify `method="lm"` where `lm` stands for linear model
- ▶ Where are also going to set `se=FALSE` (which will prevent confidence intervals from being displayed) and change the color of the regression line to red
- ▶ We can also specify the `linetype` in `geom_smooth` (i.e. solid or dashed)

## Plot the regression line

```
ggplot(florida, mapping=aes(x=Perot96, y=Buchanan00)) +  
  geom_point() +  
  xlab("Votes for Perot in 1996") +  
  ylab("Votes for Buchanan in 2000") +  
  geom_smooth(method = "lm", se = FALSE, color="red", linetype="dashed")
```

## Plot the regression line

```
## `geom_smooth()` using formula = 'y ~ x'
```



## Review: Predicted Values

Predicted values (sometimes called “fitted values”): linear regression’s estimate of what  $Y$  should be for a given  $X$

- ▶  $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 * X_i$

In our example:

- ▶  $\text{Buchanan00}_i = \hat{\beta}_0 + \hat{\beta}_1 * \text{Perot96}_i$

- ▶ Palm Beach County had 30739 votes for Perot in 1996. The linear regression would predict

$$\text{Predicted Buchanan} = 1.346 + 30739 \cdot 0.036 \approx 1108$$

## Review: Residuals

Residual: The difference between the actual and predicted

- ▶  $Y_i - \hat{Y}_i$

In our example:

- ▶  $Buchanan00_i - \hat{Buchanan00}_i$

In R we can add the residual for each county to the data frame

- ▶ This will allow us to see if Palm Beach's error is unusually large in magnitude

## Estimate regression predicting Buchanan00 with Perot96

```
lm.florida <- lm(Buchanan00 ~ Perot96, data=florida)
lm.florida

##
## Call:
## lm(formula = Buchanan00 ~ Perot96, data = florida)
##
## Coefficients:
## (Intercept)      Perot96
##      1.34575      0.03592
```

## Use names to find saved output

```
lm.florida <- lm(Buchanan00 ~ Perot96, data=florida)
names(lm.florida)
```

```
## [1] "coefficients" "residuals"      "effects"         "rank"
## [5] "fitted.values" "assign"          "qr"              "df.residual"
## [9] "xlevels"       "call"           "terms"           "model"
```

## Extract fitted values

```
florida$predBuchanan00 <- lm.florida$fitted.values
```

- ▶ This will add a variable to the data frame florida that stores the predicted values

## Extract residuals

```
florida$residuals <- lm.florida$residuals
```

- ▶ This will add a variable to the data frame `florida` that stores the residuals

## Order by largest residuals

```
arrange(florida, desc(abs(residuals)))
```

| ##    | county     | Clinton96 | Dole96 | Perot96 | Bush00 | Gore00 | Buchanan00 |
|-------|------------|-----------|--------|---------|--------|--------|------------|
| ## 1  | PalmBeach  | 230621    | 133762 | 30739   | 152846 | 268945 | 3407       |
| ## 2  | Broward    | 320736    | 142834 | 38964   | 177323 | 386561 | 788        |
| ## 3  | Lee        | 65692     | 80882  | 18389   | 106141 | 73560  | 305        |
| ## 4  | Brevard    | 80416     | 87980  | 25249   | 115185 | 97318  | 570        |
| ## 5  | Miami-Dade | 317378    | 209634 | 24722   | 289492 | 328764 | 560        |
| ## 6  | Pinellas   | 184728    | 152125 | 36990   | 184823 | 200629 | 1013       |
| ## 7  | Sarasota   | 63648     | 69198  | 14939   | 83100  | 72853  | 305        |
| ## 8  | Orange     | 105513    | 106026 | 18191   | 134517 | 140220 | 446        |
| ## 9  | Escambia   | 37768     | 60839  | 8587    | 73017  | 40943  | 502        |
| ## 10 | St.Lucie   | 36168     | 28892  | 8482    | 34705  | 41559  | 124        |
| ## 11 | Marion     | 37033     | 41397  | 11340   | 55141  | 44665  | 563        |
| ## 12 | Duval      | 112258    | 126857 | 13844   | 152098 | 107864 | 652        |
| ## 13 | Seminole   | 45051     | 59778  | 9357    | 75677  | 59174  | 194        |
| ## 14 | SantaRosa  | 10923     | 26244  | 4957    | 36274  | 12802  | 311        |

## Use pipe operator to do this

```
florida %>%  
  mutate(predBuchanan00 = lm.florida$fitted.values,  
         residuals=lm.florida$residuals) %>%  
  select(county, predBuchanan00, Buchanan00,residuals) %>%  
  arrange(desc(abs(residuals)))
```

| ##    | county     | predBuchanan00 | Buchanan00 | residuals     |
|-------|------------|----------------|------------|---------------|
| ## 1  | PalmBeach  | 1105.33818     | 3407       | 2.301662e+03  |
| ## 2  | Broward    | 1400.73939     | 788        | -6.127394e+02 |
| ## 3  | Lee        | 661.78743      | 305        | -3.567874e+02 |
| ## 4  | Brevard    | 908.16461      | 570        | -3.381646e+02 |
| ## 5  | Miami-Dade | 889.23738      | 560        | -3.292374e+02 |
| ## 6  | Pinellas   | 1329.84310     | 1013       | -3.168431e+02 |
| ## 7  | Sarasota   | 537.88054      | 305        | -2.328805e+02 |
| ## 8  | Orange     | 654.67625      | 446        | -2.086763e+02 |
| ## 9  | Escambia   | 309.74820      | 502        | 1.922518e+02  |
| ## 10 | St.Lucie   | 305.97713      | 124        | -1.819771e+02 |

## Conclusion

- ▶ The residual is largest in magnitude in Palm Beach (around 2,300 “extra” votes for Buchanan)
  - ▶ The next largest residual in magnitude is around 612 votes
- ▶ Remember: Bush only won Florida 537 votes. Quite possible that the butterfly ballot in Palm Beach altered the outcome of the election!